

Data Quality and Validation in ETL

Name: Nomit Bagaria

Assignment Name: Data Quality and Validation in ETL

Mob.No: 9828255290

Email: bagarianomit@gmail.com

SECTION A – THEORETICAL QUESTIONS

Q1. Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Answer :

Data Quality refers to the accuracy, completeness, consistency, validity, uniqueness, and reliability of data as it moves through an ETL (Extract, Transform, Load) pipeline.

It is much more than just data cleaning because it also ensures that:

- Data is complete without unnecessary missing values.
- Data follows business rules and validation checks.
- Duplicate records are removed.
- Data formats are standardized.
- Relationships between tables remain valid (referential integrity).
- The transformed data is suitable for analysis and decision-making.

Example: Simply correcting spelling mistakes in customer names is data cleaning. Ensuring every customer has a unique Customer_ID, valid transaction records, and consistent formats is part of overall data quality.

Q2. Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Answer :

Poor-quality data directly affects reports and dashboards because incorrect or incomplete information is used for analysis.

Examples include:

- Duplicate records inflate sales figures.
- Missing values produce incorrect averages.
- Invalid data formats cause calculation errors.
- Wrong customer information affects business segmentation.
- Inconsistent data leads to inaccurate KPIs.

As a result, managers may make incorrect business decisions based on unreliable insights.

Q3. What is duplicate data? Explain three causes in ETL pipelines.

Answer :

Duplicate data refers to multiple records representing the same real-world entity.

Three common causes are:

1. **Multiple Data Sources**
 - The same customer exists in two different systems.
2. **Repeated ETL Execution**
 - Running an ETL job multiple times without duplicate checks inserts the same records again.
3. **Missing Primary Key Validation**
 - Records are inserted without checking whether they already exist.

Duplicate data increases storage, reduces accuracy, and affects reporting.

Q4. Differentiate between exact, partial, and fuzzy duplicates.

Answer :

Type	Description	Example
Exact Duplicate	Every value is identical.	Rahul Mehta appears twice with identical details.
Partial Duplicate	Some values match while others differ.	Same customer ID but different phone number.
Fuzzy Duplicate	Similar but not identical values.	"Rahul Mehta" and "Rahul M."

Q5. Why should data validation be performed during transformation rather than after loading?

Answer :

Validation should occur during transformation because errors can be detected and corrected before the data reaches the destination database.

Benefits include:

- Prevents loading invalid data.
- Reduces data correction effort later.
- Maintains data integrity.
- Improves ETL performance.
- Ensures accurate reporting and analytics.

Validating before loading saves time and improves overall data quality.

Q6. Explain how business rules help in validating data accuracy. Give an example.

Answer :

Business rules are predefined conditions that data must satisfy before it is loaded into the database.

They help by:

- Checking mandatory fields.
- Enforcing valid ranges.
- Ensuring referential integrity.
- Preventing invalid or inconsistent values.

Example:

If a business rule states that **Transaction Amount must always be greater than zero**, any record with a negative amount will be rejected during the transformation stage.

Business rules ensure that only accurate and meaningful data enters the warehouse.

SECTION B – PRACTICAL QUESTIONS

Database :

```
CREATE DATABASE etl_validation_db;
```

```
USE etl_validation_db;
```

```
CREATE TABLE Sales_Transactions (
    Txn_ID INT PRIMARY KEY,
    Customer_ID VARCHAR(10),
    Customer_Name VARCHAR(50),
    Product_ID VARCHAR(10),
    Quantity INT,
    Txn_Amount INT,
    Txn_Date DATE,
    City VARCHAR(50)
);
```

```
INSERT INTO Sales_Transactions
(Txn_ID, Customer_ID, Customer_Name, Product_ID, Quantity, Txn_Amount, Txn_Date, City)
VALUES
(201,'C101','Rahul Mehta','P11',2,4000,'2025-12-01','Mumbai'),
(202,'C102','Anjali Rao','P12',1,1500,'2025-12-01','Bengaluru'),
(203,'C101','Rahul Mehta','P11',2,4000,'2025-12-01','Mumbai'),
(204,'C103','Suresh Iyer','P13',3,6000,'2025-12-02','Chennai'),
(205,'C104','Neha Singh','P14',NULL,2500,'2025-12-02','Delhi'),
(206,'C105','N/A','P15',1,NULL,'2025-12-03','Pune'),
```

```
(207,'C106','Amit Verma','P16',1,1800,NULL,'Pune'),
(208,'C101','Rahul Mehta','P11',2,4000,'2025-12-01','Mumbai');
```

```
SELECT * FROM Sales_Transactions;
```

29 • `SELECT * FROM Sales_Transactions;`
30

	Txn_ID	Customer_ID	Customer_Name	Product_ID	Quantity	Txn_Amount	Txn_Date	City
▶	201	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
	202	C102	Anjali Rao	P12	1	1500	2025-12-01	Bengaluru
	203	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
	204	C103	Suresh Iyer	P13	3	6000	2025-12-02	Chennai
	205	C104	Neha Singh	P14	NULL	2500	2025-12-02	Delhi
	206	C105	N/A	P15	1	NULL	2025-12-03	Pune
	207	C106	Amit Verma	P16	1	1800	NULL	Pune
	208	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q7. Write an SQL query on Sales_Transactions to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount).

Answer :

```
SELECT
    Customer_ID,
    Product_ID,
    Txn_Date,
    Txn_Amount,
    COUNT(*) AS Duplicate_Count
FROM Sales_Transactions
GROUP BY
    Customer_ID,
    Product_ID,
    Txn_Date,
    Txn_Amount
HAVING COUNT(*) > 1;
```

31 • `SELECT`
32 `Customer_ID,`
33 `Product_ID,`
34 `Txn_Date,`
35 `Txn_Amount,`
36 `COUNT(*) AS Duplicate_Count`
37 `FROM Sales_Transactions`
38 `GROUP BY`
39 `Customer_ID,`
40 `Product_ID,`
41 `Txn_Date,`
42 `Txn_Amount`
43 `HAVING COUNT(*) > 1;`

	Customer_ID	Product_ID	Txn_Date	Txn_Amount	Duplicate_Count
▶	C101	P11	2025-12-01	4000	3

Q8. Assume the following Customers_Master table. Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations.

Answer :

```
CREATE TABLE Customers_Master (  
    CustomerID VARCHAR(10) PRIMARY KEY,  
    CustomerName VARCHAR(50),  
    City VARCHAR(50)  
);
```

```
INSERT INTO Customers_Master  
VALUES  
( 'C101','Rahul Mehta','Mumbai'),  
( 'C102','Anjali Rao','Bengaluru'),  
( 'C103','Suresh Iyer','Chennai'),  
( 'C104','Neha Singh','Delhi');
```

```
SELECT  
    st.Customer_ID,  
    st.Customer_Name  
FROM Sales_Transactions st  
LEFT JOIN Customers_Master cm  
ON st.Customer_ID = cm.CustomerID  
WHERE cm.CustomerID IS NULL;
```

```
59 • SELECT  
60     st.Customer_ID,  
61     st.Customer_Name  
62 FROM Sales_Transactions st  
63 LEFT JOIN Customers_Master cm  
64 ON st.Customer_ID = cm.CustomerID  
65 WHERE cm.CustomerID IS NULL;  
66
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Customer_ID	Customer_Name
▶	C105	N/A
	C106	Amit Verma

